



ANDERSON JUNIOR COLLEGE
2018 JC 2 PRELIMINARY EXAMINATIONS

NAME: _____

PDG: ____/17

CHEMISTRY

Paper 2 Structured Questions

9729/02

12 September 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use		
Paper 2	1	/ 12
	2	/ 13
	3	/ 14
	4	/ 17
	5	/ 19
Total		/ 75

This document consists of **18** printed pages.

Answer **all** the questions.

- 1 (a) (i) Write an equation to show the thermal decomposition of Group 2 metal carbonate, MCO_3 .

.....[1]

- (ii) Describe and explain the trend in the thermal stabilities of the carbonates of magnesium, calcium and barium.

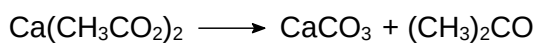
.....

[2]

- (b) MixAlco™ process is a patented technology that converts biodegradable material into mixed alcohol fuels.

This process first uses an anaerobic, mixed–culture fermentation to convert lignocellulosic biomass to carboxylate salts. These salts are then concentrated by evaporation and thermally decomposed at 450 °C into its corresponding carbonyl compounds.

The following shows the thermal decomposition of a carboxylate salt, calcium ethanoate.



- (i) Draw a dot–and–cross diagram to show the bonding in ethanoate ion, CH_3CO_2^- . In your diagram use the symbols ‘•’ and ‘x’ to distinguish electrons from the central atoms and those from the outermost atoms, and the symbol ‘□’ for any additional electrons responsible for the overall negative charge.

[1]

- (ii) Hence, suggest the shape with respect to **each** of the two carbon atoms in CH_3CO_2^- .

Shape with respect to CH_3CO_2^- :

Shape with respect to CH_3CO_2^- : [2]

- (iii) Suggest the organic product formed when a sample of $\text{Ca}(\text{CH}_3\text{CH}_2\text{CO}_2)_2$ undergoes the same decomposition.

[1]

- (iv) In one such MixAlco™ process, three organic compounds are obtained from a 1 : 1 molar mixture of $\text{Ca}(\text{CH}_3\text{CH}_2\text{CO}_2)_2$ and $\text{Ca}(\text{CH}_3\text{CO}_2)_2$.

On purification, only two of the organic compounds, **A** and **B** give yellow precipitate with aqueous alkaline iodine. The mole ratio of **A** and **B** formed is 2 : 1.

Draw the structural formulae of **A** and **B**.

A	B
----------	----------

[2]

- (c) Two samples of carboxylate salt, **C** and **D**, each containing either manganese(II) ethanoate or lead(II) ethanoate, were heated strongly.

For both samples, a solid residue and a gaseous product which gave a yellow precipitate with 2,4–dinitrophenylhydrazine were obtained. Only sample **C** gave another colourless gas which turned limewater cloudy.

By quoting suitable data from the *Data Booklet*, explain which of the samples, **C** or **D**, contains manganese(II) ethanoate.

Sample:

Explanation:

.....

.....

.....[3]

[Total: 12]

2 Use of the Data Booklet is relevant to this question.

- (a) Transition metals form complexes with different co-ordination numbers. Explain what is meant by the *co-ordination number* of a complex.

.....
[1]

- (b) Many transition metals and their complexes are paramagnetic. Paramagnetism is a property of a substance which allows it to be weakly attracted to a magnet. This property is due to the presence of unpaired electrons in the substance.

Table 2.1 below shows the relative paramagnetism of aqua complexes of vanadium in two different oxidation states.

Table 2.1

formula of complex	relative paramagnetism
$[\text{V}(\text{H}_2\text{O})_6]^{2+}$	3
$[\text{V}(\text{H}_2\text{O})_6]^{3+}$	2

- (i) Explain why transition metals can show variable oxidation states.

.....

[2]

- (ii) With reference to the electronic configurations of vanadium in the two complexes, suggest an explanation for the difference in the relative paramagnetism of the complexes in Table 2.1.

.....

[2]

- (c) Vanadium(IV) ions, VO^{2+} , are also paramagnetic.

Two reactions were carried out to investigate the changes in paramagnetism of a solution **F** containing 0.010 mol VO^{2+} ions.

Reaction 1

A sample of solution **F** is found to react completely with an acidified solution containing 0.0020 mol of KMnO_4 .

Reaction 2

Another sample of solution **F** is reduced by 0.0050 mol of $\text{Sn}^{2+}(\text{aq})$.

- (i) Using the data provided, deduce the final oxidation states of vanadium in the **Reaction 1** and **Reaction 2**. Show your working clearly.

[2]

- (ii) Hence suggest the changes in the paramagnetism of solution **F** in **Reaction 1** and **Reaction 2**. Explain your answers.

.....
.....
.....[2]

- (d) Paramagnetism is also observed in complexes of nickel. Nickel forms many complexes with a co-ordination number of 4. Complexes with a co-ordination number of 4 can take on either the tetrahedral or the square planar geometry.

Table 2.2 shows the relative paramagnetism of two nickel complexes with **different** geometry, which contain Ni^{2+} ion with a co-ordination number of 4.

Table 2.2

formula of complex	relative paramagnetism
$[\text{NiCl}_4]^{2-}$	2
$[\text{Ni}(\text{CN})_4]^{2-}$	0

Fig. 2.1 shows how the d-orbitals of a transition metal ion such as nickel are split in complexes with these two different geometry.

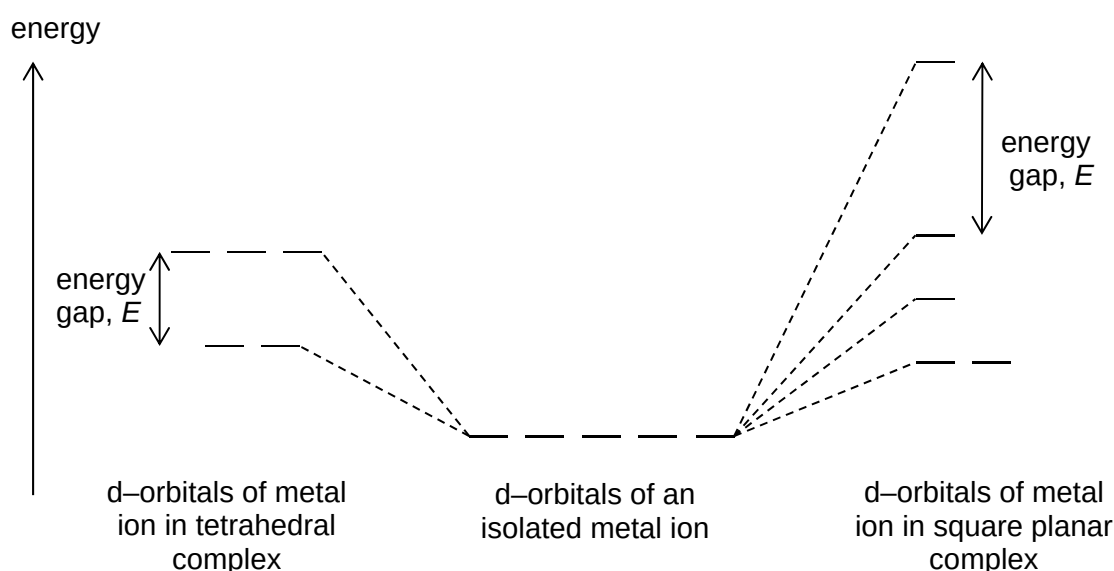


Fig. 2.1

The distribution of electrons in the d-orbitals of the transition metal ion depends on the spin state of the complex.

In a 'high spin' state, the electrons occupy all the d-orbitals singly, before starting to pair up in the lower energy d-orbitals.

In a 'low spin' state, the lower energy d-orbitals are filled first, by pairing up if necessary, before the higher energy d-orbitals are used.

The different distribution of electrons in the two spin states in the two complexes of nickel above give rise to different number of unpaired electrons, hence the observed relative paramagnetism.

Using the information and Fig. 2.1 given on page 6, deduce the geometry of the two nickel complexes and the spin state of Ni^{2+} in $[\text{Ni}(\text{CN})_4]^{2-}$.

formula of complex	geometry of complex	spin state of Ni^{2+} in complex
$[\text{NiCl}_4]^{2-}$		
$[\text{Ni}(\text{CN})_4]^{2-}$		

[2]

- (e) Nickel(II) chloride forms a 4-coordinated complex, $\text{Ni}(\text{R}_3\text{P})_2\text{Cl}_2$, with the monodentate organic ligand, triethylphosphine (where $\text{R} = \text{CH}_3\text{CH}_2-$). The complex can occur in two forms, **G** and **H**, where **G** has an overall dipole and **H** has none.

Draw the structures of **G** and **H** to account for the difference in their dipole moment.

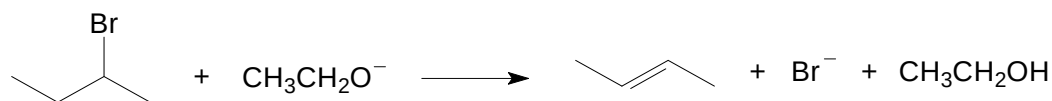
G	H
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[2]

[Total: 13]

- 3 The presence of lone pair of electrons on the oxygen atom of alkoxides (RO^-) enables them to act either as a base or a nucleophile in organic reactions.

(a) 2-bromobutane reacts with $\text{CH}_3\text{CH}_2\text{O}^-$ in ethanol according to the following equation.



- (i) Name the type of reaction that occurred.

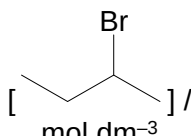
.....[1]

- (ii) Explain why $\text{CH}_3\text{CH}_2\text{O}^-$ behaves as a Lewis base in the above reaction.

.....[1]

- (iii) The kinetics of this reaction was studied and Table 3.1 contains the experimental results obtained.

Table 3.1

experiment	 / mol dm^{-3}	$[\text{CH}_3\text{CH}_2\text{O}^-] /$ mol dm^{-3}	relative initial rate
1	0.060	0.060	1.00
2	0.050	0.060	0.833
3	0.040	0.050	0.667

Use the data given above to deduce the rate law.

(iv) Hence, propose a mechanism for the above reaction given that the reaction proceeds via the following 2 steps.

- rate-determining step yields a carbocation,
- followed by acid-base reaction between H in the carbocation and $\text{CH}_3\text{CH}_2\text{O}^-$.

Describe the mechanism using the above information, including curly arrows to show the movement of electrons and all charges.

[2]

(v) Explain how the rate of reaction will be affected when 2-chlorobutane is used instead of 2-bromobutane.

.....
.....
.....[2]

- (b)** The reaction of 2-bromobutane with $\text{CH}_3\text{CH}_2\text{ONa}$ in ethanol also produced a racemic mixture of an organic by-product. The by-product has the molecular formula of $\text{C}_6\text{H}_{14}\text{O}$. It did not give white fumes when reacted with anhydrous phosphorus pentachloride.

(i) Name the type of reaction 2-bromobutane undergone.

.....[1]

(ii) With reference to the mechanism you have described in **(a)(iv)**, suggest how the racemic mixture was formed.

.....

.....

.....[1]

(iii) Draw the structures of the organic by-product present in the racemic mixture, showing their isomerism.

[2]

(iv) Quaternary carbon stereocentres are carbon atoms with four distinct carbon substituents attached. These carbon stereocentres are common features of molecules found in nature. However, production of such compounds provide a daunting challenge for chemical synthesis because a racemic mixture is often obtained.

Recently, researchers at Harvard University made use of a novel catalyst to overcome this concern. This method promotes the formation of a carbocation and binds the catalyst to one side of the carbocation.

Suggest why this method can overcome the challenge and forms only an enantiomer with specific optical activity.

.....

.....[1]

[Total: 14]

4 Both nitrogen and phosphorus are Group 15 elements.

(a) (i) Describe and explain the general trend in first ionisation energies down a group.

.....

[2]

(ii) Give the full electronic configuration of phosphorus and draw the shapes of the two types of occupied valence orbitals.

electronic configuration of P:

[2]

(b) Under standard conditions, nitrogen exists as diatomic gas, N_2 , while phosphorus occurs in one of a number of forms, including 'white' phosphorus, P_4 which contain **six** P–P single bonds.

Some of the physical properties of the compounds of nitrogen and phosphorus are also different. For example, ammonia, NH_3 , is thermally stable, whereas phosphine, PH_3 , is unstable to heat. Upon heating, gaseous PH_3 decomposes to give gaseous 'white' phosphorus, P_4 and hydrogen gas.

(i) Write the balanced equation for the thermal decomposition of one mole of PH_3 .

.....[1]

(ii) Use the bond energies given in the *Data Booklet*, determine the enthalpy change of the above reaction.

[2]

- (iii) Draw arrows on the energy diagram below to show each of the energy terms involved in the steps you have used in your calculations in (b)(ii).

Label each level with the appropriate formulae.



[2]

The thermal decomposition of phosphine is a first-order reaction. The half-life of the reaction is 35.0 s at 680 °C.

- (iv) Calculate the rate constant for this reaction and state its units.

[2]

- (v) Estimate the time taken for the gaseous phosphorus to reach 99% of its expected concentration.

[1]

- (c) Free volume, V , refers to the **volume of space** between gas molecules. For an ideal gas, free volume is essentially the same as the volume of the container and can be calculated using the ideal gas equation.

$$pV = nRT$$

- (i) Under what conditions of temperature and pressure would you expect the behaviour of a real gas to be most like that of an ideal gas?

.....[1]

- (ii) The pressure of a gaseous sample of 36.0 g of phosphine, PH_3 , in a container of volume 400 cm^3 is measured to be $1.34 \times 10^7 \text{ Pa}$ at 327°C .

Using the ideal gas equation, calculate the free volume of this gaseous sample. Express your answer in cm^3 .

[2]

- (iii) Explain, in terms of the properties of PH_3 molecules, why the volume you have calculated in (c)(ii) differs from that of the volume of the container.

.....

[2]

[Total: 17]

- 5 (a) Amino acids are the building blocks of protein in our body. Our body can synthesise certain amino acids, but others, called essential amino acids, must come from protein-containing foods in our diet. These essential amino acids are histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.

Protein is required to build, repair and maintain muscle tissue. According to dieticians, the amount of protein for any one individual depends on many factors, including their activity level, age, muscle mass, physique goals and current state of health. Table 5.1 shows the recommended daily intake for protein a sedentary person, a marathoner and a bodybuilder of the same age needs given that all other factors are the same.

Table 5.1

	Recommended Daily Intake for Protein (g / kg of body weight)
Sedentary person	0.8
Marathoner	1.4
Bodybuilder	1.8

Many athletes and bodybuilders also choose to add a protein supplement to their diets to increase their protein intake. Whey protein powder is a widely available protein supplement which typically consists 80% protein by weight with small amounts of carbohydrate and fat. The nutritional information of a particular brand of whey protein powder is shown in Fig. 5.1 below.

Serving Size – 1 scoop (25 g)

Servings Per Container – 40 (1 kg), 100 (2.5 kg), 200 (5 kg)

Nutritional Information		
	Per 100 g	Per Serving
Energy	1740 kJ	435 kJ
Fat	7.5 g	1.9 g
of which saturates	5.0 g	1.3 g
Carbohydrates	4.0 g	1.0 g
of which sugars	4.0 g	1.0 g
Protein	84 g	21 g
Salt	0.50 g	0.13 g

Fig. 5.1

- (i) State the reagents and conditions required to hydrolyse a protein non-enzymatically.

.....[1]

- (ii) Calculate the recommended daily intake of protein for a marathoner who is 65 kg.

[1]

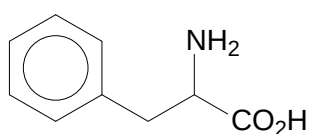
- (iii) Use information from Fig 5.1 to calculate the mass of whey protein powder that the marathoner will need to consume in a day to meet the recommended daily intake of protein.

[1]

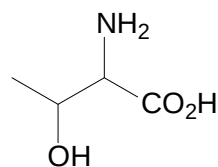
- (iv) Given that the whey protein powder contributes 20 % of the total daily energy intake of the marathoner, calculate the amount of energy contributed by his other food sources.

[1]

- (v) Suggest a simple chemical test to distinguish between phenylalanine and threonine. Describe your observations.



phenylalanine



threonine

.....

.....

.....[2]

Branched Chain Amino Acids (BCAAs) are the essential amino acids consisting of leucine, isoleucine and valine. BCAAs supplements usually offers a 2 : 1 : 1 ratio of leucine to isoleucine and valine, which is an effective combination for performance, muscle maintenance and growth benefits.

- (b) Valine, an important BCAA, can be produced using the Strecker amino–acid synthesis. This is a series of chemical reactions that converts an aldehyde into an α -amino acid. The synthesis of valine, shown in Fig. 5.2, is thought to proceed through the following stages.

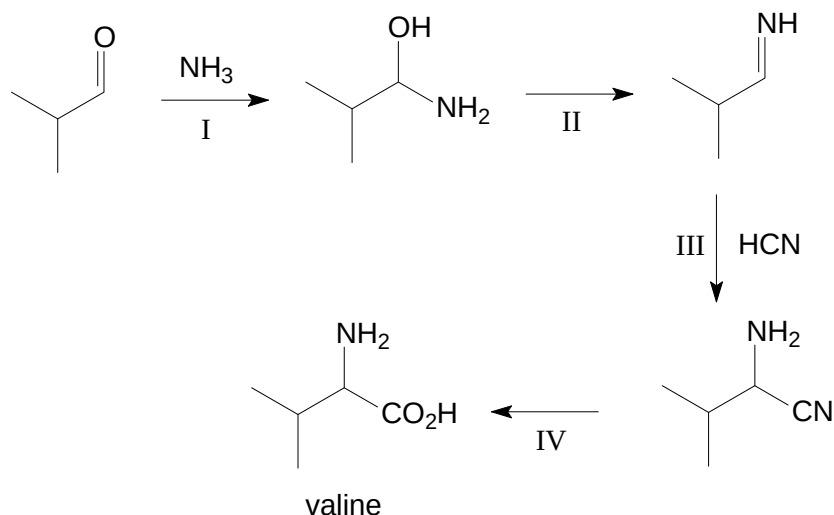


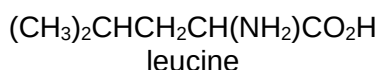
Fig. 5.2

- (i) State the type of reaction in steps I and IV.

Step I:

Step IV:[2]

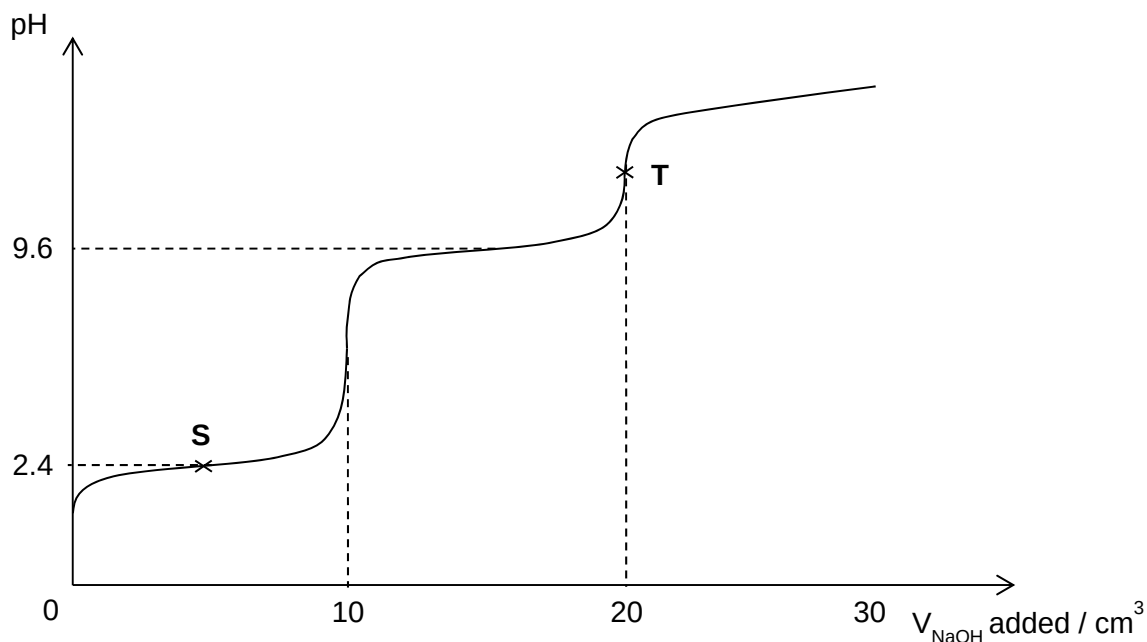
- (ii) Suggest the **skeletal** formula of the starting compound if leucine is to be synthesised via the Strecker synthesis.



[1]

- (c) There are two K_a values associated with the fully protonated form of leucine, $(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{NH}_3^+)\text{CO}_2\text{H}$.

When 10.0 cm^3 of 1.00 mol dm^{-3} $(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{NH}_3^+)\text{CO}_2\text{H}$ is titrated with 30 cm^3 of 1.00 mol dm^{-3} $\text{NaOH}(\text{aq})$, the following pH–volume added graph is obtained. You may represent the structure of protonated leucine as $\text{RCH}(\text{NH}_3^+)\text{CO}_2\text{H}$ in your answers.



- (i) Show that the K_{a2} of leucine is equal to $2.5 \times 10^{-10} \text{ mol dm}^{-3}$.

[1]

- (ii) Write an equation to explain why the pH of the solution is above 7 at point T.

.....[1]

- (iii) Hence, calculate accurately the pH of the solution at point T.

[2]

(iv) Indicate, with an "X", the isoelectric point of leucine on the graph on page 17. [1]

(v) Identify all the organic species present at point S.

[1]

(vi) Hence, with the aid of an equation, explain how the solution at S can resist a change in pH when a small amount of acid is added.

.....[1]

(vii) Determine the final pH of the solution after 30 cm³ of NaOH(aq) was added.

[1]

(d) Glutamic acid is an example of an amino acid that can be synthesised by your body and so is not essential to the human diet.

There are three pK_a values associated with glutamic acid: 2.2, 4.3 and 9.7. Starting with the fully protonated form, glutamic acid undergoes the first two stages of dissociation as shown in Fig. 5.3.

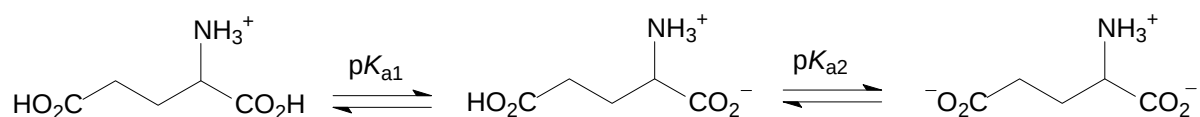


Fig. 5.3

Explain why the α -CO₂H dissociates in preference to the R group-COOH in the first stage of dissociation.

.....

[2]

[Total: 19]